

Sheet 3

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1 Assignment 7: Views and Data Manipulation

Create a view of all students in dept CSE.

```
CREATE VIEW cse_all_stud AS
SELECT *
FROM students
WHERE TRIM(deptcode) = 'CSE';
```

Create a view named as cse_stud for 'CSE' dept students having attributes rollno, name, hostel

```
CREATE VIEW cse_stud AS
SELECT rollno, name, hostel
FROM students
WHERE TRIM(deptcode) = 'CSE';
```

Insert a new student of CSE. Analyse the result.

```
INSERT INTO cse_stud (rollno, name, hostel)
VALUES ('1001', 'John Doe', 'Hostel A');
```

Observation: If the base `students` table lacks NOT NULL constraints on unlisted columns (like `deptcode` or `bdate`), this insertion will execute successfully. However, because the view does not explicitly set the `deptcode` to 'CSE' upon insertion, the new row is inserted into the base table with a NULL department. As a result, the newly inserted student will **not** appear when querying the `cse_stud` view, demonstrating the risk of inserting through views that omit filtering columns.

Increment parental income by Rs. 5000 (HRA).

```
UPDATE students
SET parent_inc = parent_inc + 5000;
```

Delete the view.

```
DROP VIEW cse_stud;  
DROP VIEW cse_all_stud;
```

Create another view of all students in dept Mechanical Engineering.

Requirement: The view will contain attributes namely Roll-No, Name, Department Name, Age.

```
CREATE VIEW mech_engineers AS  
SELECT  
    rollno AS "Roll-No",  
    name AS "Name",  
    deptcode AS "Department Name",  
    AGE(CURRENT_DATE, bdate) AS "Age"  
FROM students  
WHERE TRIM(deptcode) = 'MEC';
```

Attempt the following: Insert a new student of Mechanical Engineering Department.

```
INSERT INTO mech_engineers ("Roll-No", "Name", "Department Name", "Age"  
    )  
VALUES ('2002', 'Jane Smith', 'MEC', '20 years');
```

Observation: This insertion attempt will **fail**. PostgreSQL views containing derived or calculated columns (such as the AGE() function) are not automatically updatable. The database engine cannot reverse-engineer a date of birth (bdate) from an inserted interval ("20 years"). Insertions must be performed directly on the base `students` table.

Delete a student (for a given Name) of the same department

```
DELETE FROM mech_engineers  
WHERE "Name" = 'Tamanna Jana';
```

Observation: While some views are updatable, complex views might reject deletions without an INSTEAD OF trigger. If executed successfully, deleting from the view removes the underlying row directly from the base `students` table. This proves that views are merely virtual windows into the base table, and data destruction cascades down to the root records.

Shift a student (for a given Name) from Mechanical to Computer Science.

```
-- Shifting the student by updating the base table directly  
UPDATE students  
SET deptcode = 'CSE'  
WHERE name = 'Riddhi Agarwal' AND TRIM(deptcode) = 'MEC';
```

Observation: Changing the `deptcode` on the base table dynamically updates the views. As soon as the student's department changes to 'CSE', they automatically disappear from the `mech_engineers` view (since they no longer meet the WHERE condition) and would immediately

populate in any active 'CSE' views. This perfectly illustrates the dynamic, real-time filtering nature of SQL views.